



Title of the Final Year Undergraduate Project

An undergraduate project report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial fulfillment of the requirements for the

**Degree of the Bachelor of the Science of Engineering
Honours**

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Abstract

Automated Diagnostic Hypotheses from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decisionmaking. Current state-of-the-art (SotA) deep learning models often rely on posthoc explainability techniques that provide approximate localizations and lack direct traceability to the models' decisions. Furthermore, SotA models rarely support simultaneous multiclass classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be backprojected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

Acknowledgments

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Preface

This thesis is submitted.....

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Acronyms

SotA - State of the Art

Chapter 1

Introduction

Developing a unified pipeline for multi-class abnormality classification and grounded localization addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is accountable and its claims are supported by clinical evidence.

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality [?]. Research shows that early detection of pulmonary diseases significantly improves patient outcomes. For this, radiological imaging tools such as Chest X-Rays (CXR) and Computed Tomography (CT), plays an important role. However, the increasing volume of such imaging data increases radiologists' workload, leading to delays and potential errors in diagnosis. This has motivated the development of automated Computer-Aided Diagnosis (CAD) systems, which use computational algorithms and Artificial Intelligence (AI) to assist in tasks such as abnormality detection, classification, and segmentation [?].

Our research mainly focuses on the development of a CAD system using 3 Dimensional (3D) CT data. While CXR is commonly used as the first imaging test due to its low cost and low radiation exposure, studies show that their sensitivity to abnormalities is limited [?]. Comparatively, CT scans produce high resolution cross-sectional slices that helps detect smaller, early-stage, and subtle lung pathologies that are frequently missed by standard CXR.

However, CT analysis is complex and time-consuming, requiring radiologists to review hundreds of slices per scan, which are often inconsistent due to artifacts, anatomical variations and different scanners/protocols. Furthermore, different anatomical structures can share identical grayscale values. Telling them apart requires expert clinical judgment which is often subjective. High sensitivity can lead to detection of clinically insignificant abnormalities, while visual fatigue can lead to missed findings [?].

To address these challenges, recent research and clinical deployments of AI-based CAD systems are fundamentally changing how radiologists approach CT analysis. AI models such as 3D Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have shown to reduce interpretation time, and improve lesion detection sensitivity compared to traditional manual methods. Many of these high perform-

ing models are Deep Learning based, which are criticized for their black box nature, where their internal reasoning is difficult to interpret. This lack of interpretability is a significant barrier for clinical adoption, as radiologists require evidence to support a model’s diagnostic claims.

This has motivated the development of explainable AI (XAI) methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) that helps to visualize exactly which parts of an image a model focuses on when making a specific prediction. Such XAI methods are highly valuable for building clinical trust in deep learning models. However, they often lack the fine-grained, pixel-level precision required to identify tiny lesions, sometimes highlighting incorrect or confounding features [?].

Recent advances in sparse representation models such as Sparse Autoencoders and Dictionary Learning have shown to provide inherent interpretability in medical imaging. This stems from their ability to reconstruct complex, high-dimensional data as a linear combination of ”atoms” or features. This allows the use of efficient algorithms to easily map the output sparse domain back to the original input feature space, providing a linear relationship between the model’s decision and anatomical structures. Our work builds on these advances by applying dictionary learning to 3D Chest CT data for multi-class abnormality classification and grounded localization.

1.2 Problem Statement

The clinical utility of Deep Learning based approaches is hindered by a lack of interpretability. While tools like Grad-CAM provide post-hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model’s decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into clinical workflows, as it lacks the accountability necessary for legal and ethical compliance.

1.3 Objectives and Scope

The primary goal of this study is to address the limitations of existing AI-based approaches for abnormality classification and localization in chest CTs, by developing an inherently interpretable dictionary learning framework. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning based CAD system using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.

- Develop a unified evaluation framework for assessing the performance and interpretability of the proposed approach.
- Benchmark model performance against SotA classification and segmentation models on the same dataset.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 2 thoracic abnormalities: Pulmonary Nodules, and Ground-Glass Opacities, based on expert radiologist recommendation and dataset availability.
- This study will utilize a single publicly available dataset for training and evaluation, which may limit generalizability to other populations and scanner types.
- The proposed approach will be limited only to interpretable methods, and will not explore approaches that provide better performance for less explainability.

Chapter 2

Literature Review

2.1 Pulmonary Disease

Statistics on lung disease prevalence, morbidity, and mortality [?].

2.1.1 GGO and Pulmonary Nodules

What are pulmonary nodules and ground-glass opacities (GGOs)? Why are they clinically important? Why is early detection essential? Why are these abnormalities difficult to detect?

2.1.2 Computed Tomography (CT)

What role does CT play compared to other imaging modalities? What challenges exist in interpreting 3D chest CT scans?

2.2 Computer-Aided Diagnosis (CAD) Systems

What is Computer-Aided Diagnosis? What tasks does CAD perform? * classification
* detection * localization * segmentation

2.2.1 Evolution of CAD Systems

How have CAD systems evolved? Why has deep learning become dominant?

2.3 Deep Learning Approaches

General idea on CNNs, ViTs, and other deep learning architectures.

How well do they perform? Which datasets are commonly used? What evaluation metrics are reported?

2.3.1 nnU-Net

2.3.2 BioMedParse

BiomedParse is a foundation model for segmentation, detection, and recognition that extends prompt driven parsing to medical imaging tasks [?]. Rather than relying on a

fixed label set during inference, the model accepts a natural language prompt describing the target anatomy or pathology and produces a pixel level mask for the queried concept. This makes BiomedParse especially useful in settings where the structures of interest are heterogeneous, difficult to enumerate in advance, or described differently across institutions and reports.

In the context of chest CT, BiomedParse is attractive because many clinically relevant findings such as lung nodules, lung opacity, consolidation, and atelectasis can be queried directly with disease specific text prompts. The model combines image encoding with language understanding and uses a transformer based decoder to align the prompt with the visual features extracted from the scan [?]. The output is not only a segmentation map but also a form of interpretable localization, since the generated mask indicates where the model believes the prompted abnormality is present. Therefore, BiomedParse serves both as a SotA reference model and as a practical contrast for discussing performance, localization quality, and interpretability in 3D chest CT analysis.

2.3.3 TotalSegmentator

Accurate anatomical segmentation is an important step in CT-based computer-aided diagnosis (CAD). Earlier deep learning methods were mainly developed to segment only a few organs. To solve this problem, Wasserthal *et al.* introduced **TotalSegmentator**, a deep learning framework that can automatically segment 104 anatomical structures from a single CT scan [?]. The framework is built on the *nnU-Net* architecture, which automatically adjusts the training process to different datasets. This helps the model achieve reliable segmentation on different clinical CT images.

The model was trained using more than 1,200 routine clinical CT scans collected from different hospitals and patient groups. It achieved a mean Dice Similarity Coefficient (DSC) of about 0.94, showing high segmentation accuracy. Because of its good performance, TotalSegmentator is widely used as a preprocessing tool in medical image analysis, including computer-aided diagnosis, surgical planning, and radiotherapy planning [?].

Although TotalSegmentator provides accurate anatomical segmentation, it is not designed to classify pulmonary abnormalities such as lung nodules or ground-glass opacities. It also does not explain how its predictions are made. Therefore, additional methods are needed for disease classification and interpretable clinical decision-making.

The proposed research aims to overcome these limitations by developing an interpretable Label Consistent K-SVD (LC-KSVD) framework for pulmonary abnormality classification and grounded localization. While TotalSegmentator provides useful anatomical information, the proposed framework focuses on improving both classification performance and the interpretability of the results.

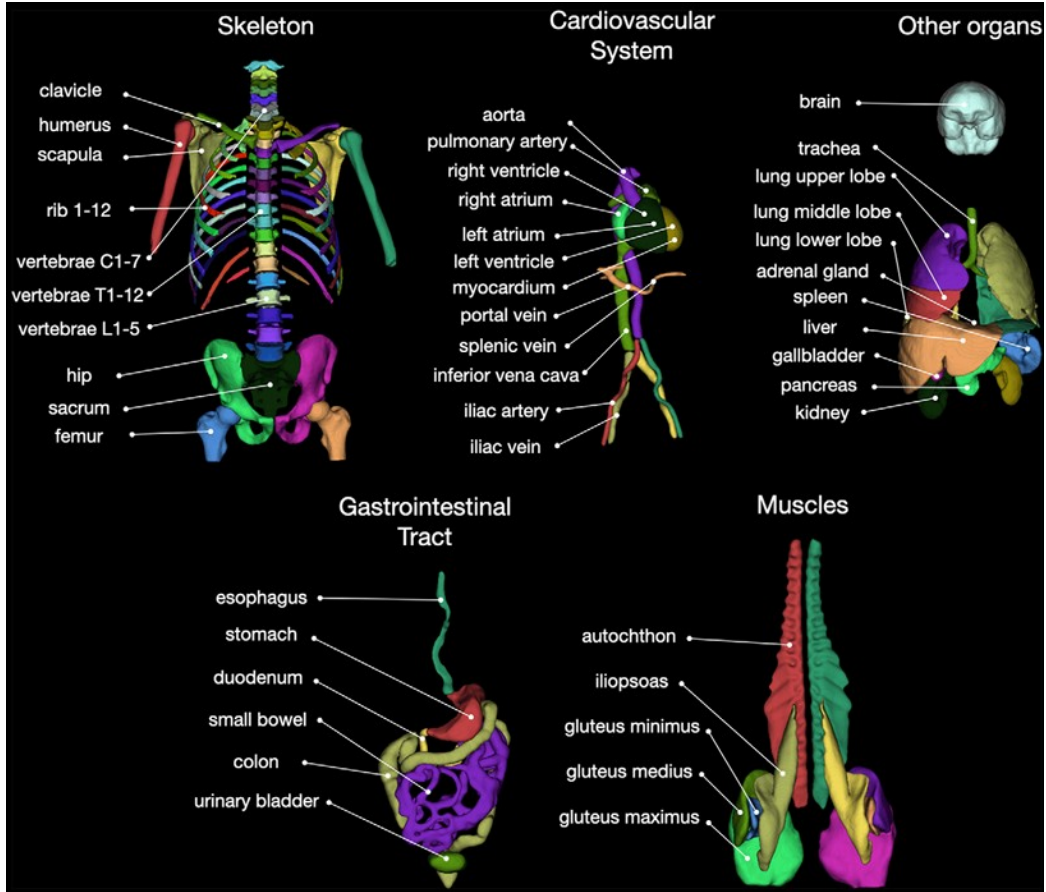


Figure 2.1: Example of automatic multi-organ segmentation produced by TotalSegmentator on a CT scan (adapted from [?]).

2.3.4 Merlin

2.3.5 MedSAM

2.3.6 CT-CLIP

CT images contain valuable information for diagnosing pulmonary diseases, while radiology reports provide detailed clinical descriptions. To learn information from both sources, Hamamci *et al.* proposed **CT-CLIP**, a vision-language foundation model for chest CT images [?]. Unlike traditional supervised models that require large amounts of labeled data, CT-CLIP learns the relationship between CT scans and their corresponding radiology reports using contrastive learning. This allows the model to learn meaningful image and text representations without task-specific annotations.

CT-CLIP consists of a 3D image encoder and a text encoder. The image encoder extracts features from CT volumes, while the text encoder processes the corresponding radiology reports. During training, the model learns to match each CT scan with its correct report in a shared feature space. The model was trained on thousands of chest CT scans paired with radiology reports, allowing it to learn both anatomical and pathological information. It has shown strong performance in pulmonary disease classification, image retrieval, and zero-shot learning tasks [?].

Although CT-CLIP achieves good performance, it mainly learns high-level feature



images/Chapter 2/ctclip_framework.png

Figure 2.2: Overview of the CT-CLIP framework showing contrastive learning between CT images and paired radiology reports (adapted from [?]).

representations instead of directly localizing pulmonary abnormalities. It also does not provide inherent explanations for its predictions, making the decision process difficult to understand.

The proposed research aims to overcome these limitations by developing an interpretable Label Consistent K-SVD (LC-KSVD) framework for pulmonary abnormality classification and grounded localization. While CT-CLIP learns rich image and text representations, the proposed framework focuses on providing transparent and interpretable diagnostic results.

2.4 Explainable AI (XAI) in Medical Imaging

Artificial intelligence (AI) has become an important tool in medical imaging by supporting tasks such as disease detection, classification, and image analysis. Many deep learning models achieve high accuracy, but their decision-making process is often difficult to understand. These models are commonly referred to as "black-box" models because they do not clearly explain how a prediction is made [? ?]. In healthcare, this lack of transparency can reduce clinicians' confidence in AI-assisted diagnosis.

Interpretability is important because medical decisions directly affect patient care.

Clinicians need to understand why a model predicts a particular disease before using its results in clinical practice. Explainable AI (XAI) provides methods that help users understand how AI models make decisions. This improves trust, supports clinical decision-making, and helps identify possible model errors or biases [?].

The importance of XAI has also increased due to ethical and regulatory requirements. Healthcare AI systems are expected to be transparent, reliable, and accountable before they are used in real clinical environments. Explainable models help meet these requirements by providing evidence that supports their predictions instead of producing results without explanation [? ?].

Explainability is especially important for chest CT diagnosis because pulmonary abnormalities such as lung nodules and ground-glass opacities can be difficult to identify. Clinicians need to know which regions of the CT scan influenced the model's prediction. This helps verify the diagnosis and increases confidence in AI-assisted clinical decisions. Therefore, improving interpretability has become an important research direction in computer-aided diagnosis systems [?].

2.4.1 SHAP

2.4.2 LIME

2.4.3 Grad-CAM

Also add Grad-CAM++

2.4.4 Other CAM Methods

Hires-CAM, Score-CAM, XGrad-CAM, etc.

2.4.5 Limitations of XAI Methods

2.5 Sparse Representation

Why is sparsity useful? How is sparse representation learned?

2.5.1 Sparse Coding

What is sparse coding? What is overcomplete representation? Why is sparse representation robust?

Two types: Greedy Algorithms and Convex Relaxation

2.5.2 Greedy Algorithms

Matching Pursuit (MP)

Orthogonal Matching Pursuit (OMP)

2.5.3 Convex Relaxation Algorithms

Least Angle Regression (LARS)

Fast Iterative Shrinkage-Thresholding Algorithm (FISTA)

2.6 Dictionary Learning

What is a dictionary atom? How is reconstruction performed? Why are atoms interpretable? What optimization algorithms exist?

2.6.1 K-Means Singular Value Decomposition (K-SVD)

2.6.2 Method of Optimized Directions (MOD)

2.6.3 Online Dictionary Learning (ODL)

2.7 Discriminative Dictionary Learning

Why isn't reconstruction enough? What is discriminative dictionary learning?

2.7.1 Label-Consistent K-SVD (LC-KSVD)

2.7.2 Fischer Discriminative Dictionary Learning (FDDL)

2.7.3 Frozen Dictionary Learning

2.8 Dictionary Learning in Medical Imaging

2.8.1 Image Reconstruction

2.8.2 Classification and Segmentation

2.9 Evaluation Metrics

2.9.1 Classification

Accuracy Precision Recall F1 ROC-AUC

2.9.2 Localization and Segmentation

Dice IoU Hausdorff Sensitivity

2.9.3 Interpretability

Faithfulness Localization accuracy Sparsity Clinical plausibility

2.10 Research Gaps

Chapter 3

Methodology

Chapter 4

Conclusion

Appendix A

Essential English Grammar - A Handout

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

A.1 Simple Present Tense

A.2 Perfect Tense

A.3 Past Past Perfect Tense

A.4 Passive Voice

A.5 Technical Writing Fundamentals

A.6 Proof Reading

Appendix B

Graphical Interpretation of Data

The graphical interpretation given below shows the relationship between x and y is acceptable to the purpose of the project

B.1 Data Analysis

B.1.1 Statistical Analysis

B.1.2 Empirical Analysis

B.2 The X vs Y vs Z